ATHARVA.EXE: Game Documentation (PS01 GlitchVerse)

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1. Project at a glance

ATHARVA.EXE is a browser-based 2D puzzle-platformer where the player edits the broken rules of a glitching college campus (gravity, time and reality) to reach the Server Room and reboot the system. It is our entry for PS01: GlitchVerse at Artificial Arcade VI, Atharva University, Mumbai.

Item	Detail
Game title	ATHARVA.EXE
Event	Artificial Arcade VI, Atharva University, Mumbai (24-hour build)
Problem statement	PS01: GlitchVerse, "The World Is Broken"
Category	Single-player puzzle and adventure
Platform	Web browser (desktop, keyboard controls), hosted online
Engine	Phaser 3 with Vite
Levels	4 main levels (Main Gate, Library, Canteen and Corridors, Server Room)
Environment-altering mechanics	4 (Gravity Flip, Time Freeze, Dimension Switch, Time Echo)
Team name	To be filled in by the team

What makes it unique

1. **The rules are visible.** A live "Rules Panel" shows the game's own variables (gravity, time_scale, layer) and the player edits them to solve puzzles. The theme is obvious in the first 10 seconds.

2. **It is set on a campus.** Levels are college spots every judge and student recognises, and the humour (proxy attendance, CGPA scoring, exam bell) is campus humour.
3. **Four mechanics that combine.** Each level teaches one rule; the final level needs all of them together.
4. **Time Echo.** A ghost clone replays the player's last 5 seconds and holds down pressure plates, a mechanic few teams will attempt.
5. **Glitch Energy.** Every rule change costs energy, so the player must plan, not spam.
6. **Event-matched look.** The neon sunset palette is taken from the Artificial Arcade VI poster.

2. Story and concept

The campus's central computer has crashed, and the player, a student-debugger, must fix it from the inside before the exam-bell timer runs out.

The story in short: A bug has corrupted the campus system. Classrooms float, clocks run backwards and two versions of the campus (Normal and Glitch) are mixed together. The only way to move forward is to change the world's rules. The player fights through the Main Gate, Library and Canteen to reach the Server Room, where the "Corrupted Admin" tries to stop the reboot.

The core idea: the Rules Panel. A small panel on the screen shows the live state of the world like code:

```
gravity    = DOWN
time_scale = 1.0
layer      = NORMAL
```

When the player presses a power key, the panel changes in front of them (for example `gravity = UP`). Judges see at once that the player is editing the game's own code. This is the pitch in one image.

Tone: funny, neon, affectionate towards college life. The Corrupted Admin is a fictional system, not a real person.

Theme fit (PS01): the world is broken (glitches), and the player manipulates its fundamental rules (gravity, time, dimension) to solve puzzles, which is exactly what GlitchVerse asks for.

3. How the game works (player's view)

The player's goal is to clear 4 levels, reach the exit door in each, and reboot the campus system in the Server Room, scoring as high as possible on the way.

Controls

Key	Action	Costs Glitch Energy?
Left / Right arrows or A / D	Move	No
Space or Up arrow	Jump	No
G	Flip gravity (fall up or down)	Yes
F	Freeze time	Yes
Q	Switch dimension (Normal or Glitch campus)	Yes
E	Record and replay the Time Echo ghost (unlocked in Level 4)	Yes
R	Instant restart of the current level	No
Esc	Pause menu	No

Resources and health

Meter	What it is	Rules
Glitch Energy	Fuel for every power, shown as a bar at the top	Starts full (100). Each power costs energy. Refills slowly (about 8 per second) while the player stands on solid ground.
Attendance %	The player's health	Starts at 100%. Lasers, spikes and falling objects remove 25% per hit. At 0% the "DETAINED" screen appears and the level restarts.

Meter	What it is	Rules
Exam Bell timer	The level countdown	Each level has a time limit. If it hits zero the level is lost and restarts. Finishing early gives bonus points.

Collectibles

Item	Effect
Data bit (small glowing cube)	+10 points
Coffee	Short speed boost and a small energy refill
ID card	Opens a locked door in the level
Notes (rare, hidden)	+50 bonus points; hunting for them rewards exploration

A level from start to finish

1. Read the one-line hint shown at the start of the level.
2. Look at the room and spot the problem (a high exit, moving lasers, a wall).
3. Use the right power at the right moment, watching the energy bar.
4. Collect data bits on the way if it is safe.
5. Reach the exit door. The level-complete screen shows score, time and stars.
6. Press the next-level button or retry for more stars.

4. The four environment-altering mechanics

PS01 requires at least 3 mechanics that change the environment; we build 4, each with a clear rule, a cost and a way to build it.

Mechanic	Key	Energy cost	Duration / cooldown	What the player can do
Gravity Flip	G	20	Instant toggle, 0.5 s cooldown	Fall up to ceilings and reach platforms that are only reachable upside down

Mechanic	Key	Energy cost	Duration / cooldown	What the player can do
Time Freeze	F	35	Lasts 3 s, 6 s cooldown	Stop moving hazards and use them as stepping stones
Dimension Switch	Q	15	Instant toggle, 1 s cooldown	Swap between the Normal and Glitch campus, where walls, platforms and hazards differ
Time Echo	E	40	Records 5 s, then replays	A ghost clone repeats the recorded moves and holds pressure plates

These numbers are starting values. Tune them while playtesting.

4.1 Gravity Flip

- **Rule:** pressing G reverses gravity for the player. The player sprite flips upside down and Left/Right still move the same way on screen.
- **Teaches:** the world's vertical rules are not fixed.
- **How to build:** keep a `gravityDir` variable (+1 or -1). Set the player's vertical gravity to `gravityDir * 900`, flip the sprite with `setFlipY`, and invert the jump direction. Landing detection must check `body.blocked.up` when gravity is up.
- **Watch out for:** the jump, coyote-time and ground checks must all use the current gravity direction.

4.2 Time Freeze

- **Rule:** pressing F freezes every moving hazard (lasers, falling books, sliding platforms) for 3 seconds. The player and the Rules Panel keep working. The screen tints blue and a ticking sound plays while time is frozen.
- **Teaches:** danger can become a safe path.
- **How to build:** every hazard moves using a shared `hazardTimeScale` variable (1 normally, 0 when frozen). Do not use the physics world's own time scale, because that would also freeze the player.
- **Watch out for:** a frozen platform must still carry or support the player, so keep its collider active.

4.3 Dimension Switch

- **Rule:** the campus exists in two layers, Normal and Glitch. Each platform, wall and hazard belongs to one layer or to both. Pressing Q makes the other layer solid and fades the current one to a faint outline.
- **Teaches:** some things exist in only one world.
- **How to build:** make one physics group per layer. On switch, enable the bodies of the active group, disable the others and change their alpha. Use a flash and a short glitch shader pulse to hide the swap.
- **Watch out for:** if the player switches while standing inside a block of the other layer, push them out to the nearest free space (or block the switch and show "UNSAFE").

4.4 Time Echo (our standout mechanic)

- **Rule:** the first press of E starts recording the player's moves. The second press (or after 5 s) ends the recording, and a ghost clone replays it in a loop. The ghost cannot be hurt but can press pressure plates.
- **Teaches:** the player can cooperate with their own past.
- **How to build:** each frame, push `{x, y, flipY, gravityDir}` into an array. On replay, move the ghost sprite to the next stored position each frame. Check overlap between the ghost and pressure plates the same way as the player.
- **Watch out for:** record at a fixed rate (for example 60 samples a second) so the replay is exact even if the frame rate dips.

How the mechanics combine

Combination	Result
Gravity Flip + Time Freeze	Freeze a falling book mid-air, flip, and walk on it upside down
Dimension Switch + Gravity Flip	Reach a ceiling platform that only exists in the Glitch layer
Time Echo + Dimension Switch	The ghost holds a plate in one layer while the player crosses the other
All four	The final Server Room puzzle

5. Level-by-level design

Every level follows one pattern: teach one rule safely, test it, then add a twist. Difficulty rises each level and the last level needs every rule.

Level	Location	New mechanic	Time limit	Difficulty	Puzzles in the level
1	Main Gate	Gravity Flip	90 s	Easy	3
2	Library	Time Freeze	120 s	Medium	4
3	Canteen and Corridors	Dimension Switch	150 s	Medium-hard	4
4	Server Room	Time Echo + all others	180 s	Hard	5

The layouts below are design targets. Exact positions are decided while building and playtesting.

Level 1: Main Gate (learn Gravity Flip)

- **Goal:** reach the ID-card scanner and open the gate.
- **Intro hint:** "The gate is above you. Gravity is just a setting. Press G."
- **Obstacles:** spike strips on the floor, a gap too wide to jump, and an exit placed on the ceiling.
- **Puzzle 1 (tutorial):** a short corridor with the exit on the ceiling. Press G, fall upward and walk to the exit side.
- **Puzzle 2:** a floor with spikes. Flip gravity to walk along the ceiling and skip the spikes.
- **Puzzle 3:** flip gravity mid-jump over a gap to land on the ceiling platform, then flip back to land at the scanner.
- **Collectibles:** 8 data bits (2 only reachable on the ceiling), 1 coffee, 1 hidden note.
- **Safe to fail:** spikes only cost 25% attendance and checkpoints are close together.
- **Stars:** 1 star for finishing, 2 stars for finishing under 60 s, 3 stars for finishing under 60 s with at least 6 data bits.

Level 2: Library (learn Time Freeze)

- **Goal:** cross the floating bookshelves and reach the Librarian's terminal.
- **Intro hint:** "Everything is moving. Stop the clock. Press F."
- **Obstacles:** books that fall from shelves on a loop, laser scanners sweeping left to right, and bookshelf platforms that slide up and down.

- **Puzzle 1:** a laser sweeps the full width of the floor. Freeze it when its beam is at the far edge and run under.
- **Puzzle 2:** falling books create a staircase. Freeze while they are mid-air, jump from book to book to reach a high shelf.
- **Puzzle 3:** a sliding bookshelf platform moves too fast to board. Freeze it next to the player and step on.
- **Puzzle 4 (combo, uses Level 1 skill):** a book frozen mid-fall is positioned above the exit. Flip gravity and walk on its underside.
- **Energy lesson:** Time Freeze costs 35, so the player can only freeze about twice in a row. They must plan which puzzle needs it most.
- **Collectibles:** 10 data bits (several on frozen-only paths), 2 coffees, 1 ID card (opens the shelf gate), 1 hidden note.
- **Stars:** 1 star for finishing, 2 stars for finishing under 90 s, 3 stars for finishing under 90 s using at most 3 powers.

Level 3: Canteen and Corridors (learn Dimension Switch)

- **Goal:** reach the canteen kitchen door through a maze that changes with the layer.
- **Intro hint:** "There are two campuses here. Press Q to step into the other one."
- **The two layers:** the Normal campus has solid walls and clean floors. The Glitch campus has different walls, floating tables and corrupted (red) hazards.
- **Puzzle 1:** a wall blocks the path in Normal. In Glitch it does not exist. Switch, walk through, switch back.
- **Puzzle 2:** a gap has a bridge only in Glitch. Cross while in Glitch, but the landing zone has a Glitch-only laser, so switch back right after crossing.
- **Puzzle 3:** a moving trolley (hazard) exists only in Normal. Switch to Glitch to avoid it, then switch back to use a Normal-only platform.
- **Puzzle 4 (combo):** a ceiling corridor exists only in Glitch. Combine Gravity Flip and Dimension Switch to travel along it.
- **Collectibles:** 12 data bits (in both layers), 2 coffees, 1 ID card, 1 hidden note.
- **Stars:** 1 star for finishing, 2 stars for finishing under 110 s, 3 stars for finishing under 110 s with 8 or more data bits.

Level 4: Server Room (final level, all mechanics and Time Echo)

- **Goal:** reach the Root Terminal and press the REBOOT button, then survive the Corrupted Admin's last attack.
- **Intro hint:** "The Admin controls the rules now. Take them back."
- **New mechanic: Time Echo (E).** Introduced in the first puzzle with a safe tutorial plate.
- **Puzzle 1 (Echo tutorial):** a door opens only while a pressure plate is held. Record a short walk onto the plate, let the ghost hold it, and walk through the door.
- **Puzzle 2 (Echo + Dimension):** a plate exists only in the Glitch layer. The ghost holds it in Glitch while the player stays in Normal to use a Normal-only bridge.
- **Puzzle 3 (Freeze + Gravity):** server racks fall in a loop. Freeze them, flip gravity and walk across their undersides to the next zone.
- **Puzzle 4 (energy management):** a hallway needs all three powers in sequence, but the energy bar only allows two at once. The player must wait for recharge or find the coffee.
- **Puzzle 5 (final):** the Root Terminal sits behind two doors and a laser. This needs an Echo plate, a freeze and a gravity flip in one run.
- **The Corrupted Admin twist:** every 20 seconds the Admin triggers a random glitch (a forced gravity flip, a 2-second freeze of the player's controls, or a forced layer switch). A warning appears for 2 seconds first, for example "WARNING: GRAVITY OVERRIDE INCOMING", so the player can prepare. It is fair, not random cheating.
- **Collectibles:** 15 data bits, 3 coffees, 1 ID card, 1 hidden note.
- **Stars:** 1 star for finishing, 2 stars for finishing under 140 s, 3 stars for finishing under 140 s with at most 6 powers used.

Optional bonus levels (only if time remains)

Computer Lab, Auditorium and Ground can reuse the same mechanics with new layouts. Do not start these until all four main levels are polished.

Difficulty rules for level design

1. Never kill the player on the first sight of a new hazard. Show it, then test.
2. Restart must take under 1 second and put the player back near the puzzle, not at the start of the level.
3. A level must be solvable with the energy the player has at that point.
4. Every puzzle has one clear solution and may have a faster expert route.

5. Playtest each level with a teammate who has not seen it.

6. Scoring system

Each level score combines collected items, speed and smart use of powers, and the total becomes a CGPA out of 10 with a campus-style rank.

Points per level

Source	Points
Data bit	+10 each
Hidden note	+50
Time bonus	+2 per second left on the Exam Bell timer
Efficiency bonus	+25 for each power not used below a budget of 6 powers (minimum 0)
Death penalty	-20 per DETAINED screen
Level clear	+100

These values are starting points. Adjust them so a skilled run lands near the top of the scale and a careless run near the bottom.

Stars

Each level awards 1 to 3 stars based on the targets in section 5. The stars show on the level-select screen so players want to replay.

CGPA and ranks

CGPA = (total score of all levels / maximum possible score) x 10, rounded to one decimal. The maximum possible score is measured once by a perfect run and stored as a constant.

CGPA	Rank
Below 4.0	Backbencher
4.0 to 6.9	Average Student

CGPA	Rank
7.0 to 8.9	Topper
9.0 to 10	Dean's List

Saving scores

- Best score and stars per level are saved in the browser's `localStorage`.
- The final results screen shows the total score, CGPA, rank, total time, deaths and powers used.
- A "Reset progress" option on the menu clears saved data (useful before the live demo).

7. Screens, flow, win and loss conditions

The game has eight screens, and every PS01 requirement for progression, restart and clear win/loss conditions is handled by them.

Screen	Purpose	Key elements
Loading	Preload assets	Glitch-styled progress bar, "Artificial Arcade VI presents"
Main menu	Entry point	Play, Level Select, How to Play, Reset Progress, sound toggle
How to Play	Teach controls	Key map, the four powers, energy and attendance explained in one page
Level select	Choose a level	4 level cards with stars and best score; locked levels are greyed out
Game (HUD)	The playable level	Rules Panel, Glitch Energy bar, Attendance %, timer, score, pause button
Pause	Break	Resume, Restart level, Quit to menu
Level complete	Result of one level	Score breakdown, stars, time, buttons for Next level and Retry
Final results	End of the game	Total score, CGPA, rank, deaths, best times, Play again

Level progression

1. Level 1 is open from the start.
2. Completing a level unlocks the next one.
3. Progress is saved so the player can return later.
4. Completing Level 4 shows the Final results screen.

Win and loss conditions

Outcome	Trigger	What happens
Level win	Player touches the exit	Level complete screen with score and stars
Game win	Player presses REBOOT in Level 4	Reboot animation, then the Final results screen
Loss: health	Attendance reaches 0%	"DETAINED" glitch screen with a fake error message, then instant respawn
Loss: time	Exam Bell timer reaches 0	"EXAM BELL RANG" screen, then restart of the level
Loss: fall	Player leaves the world bounds	Counts as a hit, respawn at the last checkpoint

Restart system

- Press R at any time to restart the current level (the score for that attempt resets).
- Death respawns the player at the nearest checkpoint within about 0.5 to 1 second.
- The pause menu offers Restart level and Quit to menu.

Fun death messages keep failure light: "SEGMENTATION FAULT", "NullPointerException", "404: ATTENDANCE NOT FOUND", "DETAINED".

Screen flow

```
Loading -> Main menu -> Level select -> Game level -> Level complete -> (next level or Final results)
                    \-> How to Play        \-> Pause / Death -> Restart
```

8. Visual and audio style

The look is neon-glitch on a dark base, using the sunset palette of the Artificial Arcade VI poster, so the game reads as part of the event while needing almost no hand-drawn art.

Palette

Role	Colour	Hex
Background	Deep navy	#0B1026
Primary glow	Cyan	#00E5FF
Glitch / danger	Magenta pink	#FF2E93
Warm accent	Sunset orange	#FF9E3D
Highlight / collectibles	Warning yellow	#FFE14D
UI panels	Purple	#7B4BFF

Use these six colours everywhere. A consistent palette is the cheapest way to look professional.

Visual techniques

- **Shapes drawn in code or SVG** for the player, platforms, hazards and UI. They stay sharp at any screen size.
- **Bloom and glow** on outlines, collectibles and lasers.
- **Parallax background** with 2 to 3 layers (blurred, purple-tinted campus photos and silhouettes) drifting at different speeds.
- **Glitch shader** (chromatic aberration, scanlines, slight screen tear) that grows stronger in later levels and on every power use.
- **Particles** for jumps, landings, collecting, deaths and each glitch activation.
- **Layer styling:** the Normal campus is clean and cool-toned; the Glitch campus is magenta, distorted and flickering.
- **Freeze look:** a blue tint and a frosted outline on frozen objects.
- **Echo look:** the ghost is a translucent cyan copy of the player with a trailing afterimage.

Campus feel

- Use real campus photos only as blurred, tinted backgrounds, never as raw photos.
- Put the university name on signboards in the world and a short "Welcome to Atharva" glitch intro.
- Use the event name on the loading screen.
- Keep humour friendly. No real person (a principal, chancellor or teacher) appears as a character or joke. The antagonist is the fictional "Corrupted Admin System".
- Do not copy the poster's characters or logo. Only its colour mood is borrowed.

Audio plan

Sound	Use
Jump, land, collect, hit	Short retro effects generated with a free sound tool
Gravity flip	Rising whoosh
Time freeze	Low ticking then a freeze chime
Dimension switch	Glitchy static burst
Echo record / replay	Tape-rewind blip
Death	Error buzz
Level complete	Short victory jingle
Music	One calm loop for menu, one faster loop for levels, a tense layer in Level 4

Use only free or royalty-free audio and credit it on the results screen.

9. Tech stack and architecture

We build the game with Phaser 3 and Vite because it gives a web team physics, input, audio, cameras, particles and WebGL effects out of the box, so we only write the game logic.

Need	Choice	Why
Game engine	Phaser 3	2D engine for the web, WebGL rendering, large community
Build tool	Vite	Fast dev server and a production build
Language	JavaScript (or TypeScript if the team is comfortable)	Matches the team's web skills
Physics	Phaser Arcade Physics	Fast and simple; enough for platforming
Levels	Hand-written tile arrays, or the Tiled editor	Tile arrays are the quickest to start
Effects	Phaser built-in FX plus a small custom shader	Glow, bloom, glitch look
Sound	jsfxr or ChipTone, plus free music	Quick retro sound effects
Assets	Kenney.nl, OpenGameArt, itch.io free packs	Free licences
Hosting	Vercel or Netlify	Live link for the judges

Phaser's effects API has changed between versions, so check the documentation of the installed version before using a specific effect.

Project structure

```
atharva-exe/
  index.html
  package.json
  vite.config.js
  public/assets/        images, audio, backgrounds
  src/
    main.js             Phaser game config
    config.js           constants: energy costs, time limits, colours
    scenes/
      BootScene.js      loading bar
      MenuScene.js
      HowToPlayScene.js
```

```
    LevelSelectScene.js
    GameScene.js      runs any level
    UIScene.js        HUD (runs on top of GameScene)
    ResultsScene.js
  entities/
    Player.js         movement, coyote time, jump buffer, powers
    Ghost.js          Time Echo clone
    Hazard.js         lasers, falling books, spikes
    Plate.js          pressure plates and doors
  systems/
    RulesState.js     gravity, timeScale, layer, energy
    Score.js          points, stars, CGPA, localStorage
    Effects.js        shaders, screen shake, particles
    Audio.js
  levels/
    level1.js  level2.js  level3.js  level4.js
```

Key design decisions

1. **One shared `RulesState`.** A single object holds `gravityDir`, `hazardTimeScale`, `layer` and `energy`. Every system reads from it, and the Rules Panel on screen is drawn from it, so the visible "code" always matches the real game state.
2. **One `GameScene` for all levels.** Each level file is just data (a tile map, spawn point, exit, hazards, plates, time limit). Adding a level means adding a file, not new logic.
3. **A separate `UIScene`** draws the HUD on top, so camera shake and shaders do not distort the interface.
4. **Fixed-step recording for Time Echo** so the replay is exact.
5. **Level data as plain objects** so the team can design levels in parallel.

Level data shape (example)

```
{
  id: 1,
  name: "Main Gate",
  timeLimit: 90,
  spawn: { x: 64, y: 480 },
  exit: { x: 1700, y: 80 },
  allowedPowers: ["gravity"],
  tiles: [ ... ],
  hazards: [ { type: "spikes", x: 400, y: 540 } ],
  collectibles: [ { type: "bit", x: 300, y: 450 } ],
  starTargets: { two: 60, three: { time: 60, bits: 6 } }
}
```

10. Performance and game feel

The target is a steady 60 frames per second on an average laptop, with controls that feel instant and forgiving.

No-lag checklist

- [ ] Use `Phaser.AUTO` (WebGL) with `powerPreference: 'high-performance'`.
- [ ] Pack sprites into one texture atlas to reduce draw calls.
- [ ] Use object pools for particles, lasers and falling books; never create objects inside `update()`.
- [ ] Keep dynamic physics bodies few; use static groups for tiles.
- [ ] Preload every asset in the loading scene so nothing loads mid-level.
- [ ] Keep background images small (compressed, under about 1 MB each).
- [ ] Limit the shader to one full-screen pass.
- [ ] Test the production build (`vite build` then preview), not the dev server.
- [ ] Check frame rate in the browser's Performance tab on the weakest laptop in the team.
- [ ] Provide a "Low effects" toggle that turns off the shader and reduces particles.

Game feel checklist

Technique	What it does	Starting value
Coyote time	Player can still jump just after leaving a ledge	100 ms
Jump buffer	A jump pressed slightly early still counts	120 ms
Variable jump height	Releasing the key early gives a shorter jump	Cut upward speed by half on release
Camera follow with lerp	Smooth camera instead of a rigid lock	0.1
Screen shake	Impact feedback on hits and flips	80 to 150 ms, small amplitude
Hit-stop	Tiny freeze on death or big events	50 ms
Squash and stretch	Player sprite stretches on jump and squashes on landing	10 to 20% scale change
Acceleration and friction	Movement ramps up and slows smoothly	Tune by feel
Sound on every action	Confirms every input	See audio plan
Screen transitions	Glitch wipes and fades between scenes	300 to 500 ms

Playtest the controls first. If the movement does not feel good in an empty room, no level design will fix it.

11. PS01 requirements checklist

Every compulsory requirement from the PS01 brief maps to a feature in this document, and the team ticks each one before submitting.

PS01 compulsory requirement	How ATHARVA.EXE meets it	Section	Done
Minimum 4 playable levels	Main Gate, Library, Canteen and Corridors, Server Room	5	To do
At least 3 unique environment-altering mechanics	Gravity Flip, Time Freeze, Dimension Switch, plus Time Echo	4	To do
Interactive puzzles or obstacles in every level	3 to 5 puzzles per level, plus lasers, spikes, falling objects and plates	5	To do
Functional level progression and restart system	Level unlocks, level select, R to restart, checkpoints	7	To do
Scoring or timer-based performance tracking	Points, Exam Bell timer, stars, CGPA	6	To do
Clear win/loss conditions and user-friendly interface	Win and loss table, clear HUD, How to Play page, pause menu	7	To do

PS01 demonstration point	How we show it	Done
Creative environment manipulation	Rules Panel with gravity, time and layer changes	To do
Interactive puzzles and obstacles	Puzzles listed per level	To do
Progressively challenging levels	Difficulty table in section 5	To do
Smooth player movement and controls	Game feel checklist in section 10	To do
Engaging gameplay with a scoring system	Scoring, stars, CGPA and ranks	To do

Judging area from the brief	Our answer
Creativity and innovation in game mechanics	Time Echo and the visible Rules Panel
Game logic and level design	Teach, test, twist pattern in every level

Judging area from the brief	Our answer
Problem-solving and puzzle development	Combined-power puzzles and energy management
Player engagement and user experience	Campus humour, instant restart, CGPA ranks
Smooth gameplay and visual presentation	60 FPS target, neon glitch visuals, sound

12. Team roles and 24-hour timeline

The build is split into five phases, and the first rule is to get one level fully playable before expanding. Roles below assume a team of four; merge or split them to match the real team size.

Role	Owns	Main files
Gameplay lead	Player movement, the four mechanics, RulesState	Player.js, Ghost.js, RulesState.js
Level designer	Level layouts, puzzles, difficulty tuning, playtesting	levels/, Hazard.js, Plate.js
UI and visuals	Menus, HUD, Rules Panel, shaders, particles, palette	UIScene.js, Effects.js, scenes/
Audio, scoring and deploy	Sounds, music, score and CGPA, localStorage, hosting, demo video	Score.js, Audio.js

Timeline

Hours	Phase	Goal	Done when
0 to 3	Setup	Phaser and Vite project running, player moves and jumps with good feel, camera follows, one test room	Player controls feel smooth

Hours	Phase	Goal	Done when
3 to 8	Mechanics	Gravity Flip, Time Freeze, Dimension Switch each working in its own small test room, energy bar and Rules Panel live	Each power works and shows on the panel
8 to 14	Levels and flow	Levels 1 to 4 built, menu, level select, HUD, scoring, restart, win and loss screens, Time Echo added	Game can be played start to end
14 to 19	Polish	Shader, particles, parallax, sounds, music, transitions, death messages	Looks and sounds finished
19 to 22	Testing	Playtest with outsiders, fix bugs, tune difficulty, add hints, low-effects toggle	No crashes, every level beatable
22 to 24	Delivery	Production build deployed, backup demo video recorded, pitch rehearsed, final checklist ticked	Live link works on another device

Working rules

1. Commit to a shared Git repository every hour so nothing is lost.
2. Do not add a new feature after hour 19; only fix and polish.
3. Cut scope early: if a level or power is not working by hour 14, simplify it.
4. One person owns the final build and deploy; nobody else pushes to the live branch in the last 2 hours.

13. Risks, testing, deployment and pitch

The biggest risk is trying to build too much, so the fallback plan is to ship four polished levels with three mechanics and add Time Echo only if time allows.

Risks and fallbacks

Risk	Likelihood	Fallback
Team has no game-dev experience, so physics bugs eat time	High	Use Phaser Arcade Physics only, keep levels simple, build and test each mechanic in a small room first
Time Echo is too hard to finish	Medium	Replace it with a simpler fourth idea (for example a moving-platform switch); the three core mechanics already meet the requirement
Dimension Switch causes the player to get stuck in walls	Medium	Block the switch when unsafe and show "UNSAFE", or push the player to the nearest free tile
Lag on judges' machine	Medium	Low-effects toggle, small assets, test the production build
Level too hard or unclear	High	Playtest with someone outside the team, add one-line hints
Deployment fails at the last minute	Medium	Deploy a first version in the first 6 hours, keep a local copy and a backup demo video
Running out of time	High	Feature freeze at hour 19

Testing checklist

- [ ] Every level can be completed from start to finish
- [ ] Every level can be failed and restarted without a bug
- [ ] Each power works in each level that allows it, and is blocked where it should be
- [ ] Energy cannot go below 0 and refills correctly
- [ ] Time Freeze never freezes the player's own controls
- [ ] Dimension Switch never traps the player inside a wall
- [ ] Level progression unlocks the next level and saves
- [ ] Final results screen shows the correct score, CGPA and rank
- [ ] Works in Chrome and Edge at 1366x768 and 1920x1080
- [ ] Keyboard focus is on the game when the page loads

- [ ] Sound can be muted, and the game works without sound
- [ ] Runs at about 60 FPS on the weakest laptop

Deployment

1. Run the production build and test it locally.
2. Deploy to Vercel or Netlify and open the link on a different device.
3. Keep the link, a zipped copy of the project and a screen recording of a full playthrough ready for the judges.

Demo and pitch (about 3 minutes)

1. **Hook (20 s):** open on the Rules Panel and flip a rule live. Say: "The player edits the game's own code to survive."
2. **Problem and idea (20 s):** one sentence on PS01 and our campus-themed answer.
3. **Live play (90 s):** Level 1 gravity flip, Level 2 time freeze, then jump to Level 4 and show the Time Echo ghost holding a plate.
4. **Scoring and polish (20 s):** show stars, CGPA and rank, mention the event-matched look.
5. **Close (10 s):** "We did not build a game about a college. We built a game where the college's own rules are broken, and you fix them by editing the rules."

Tip: load the game with all levels unlocked for the demo, so the presenter can jump to any level.